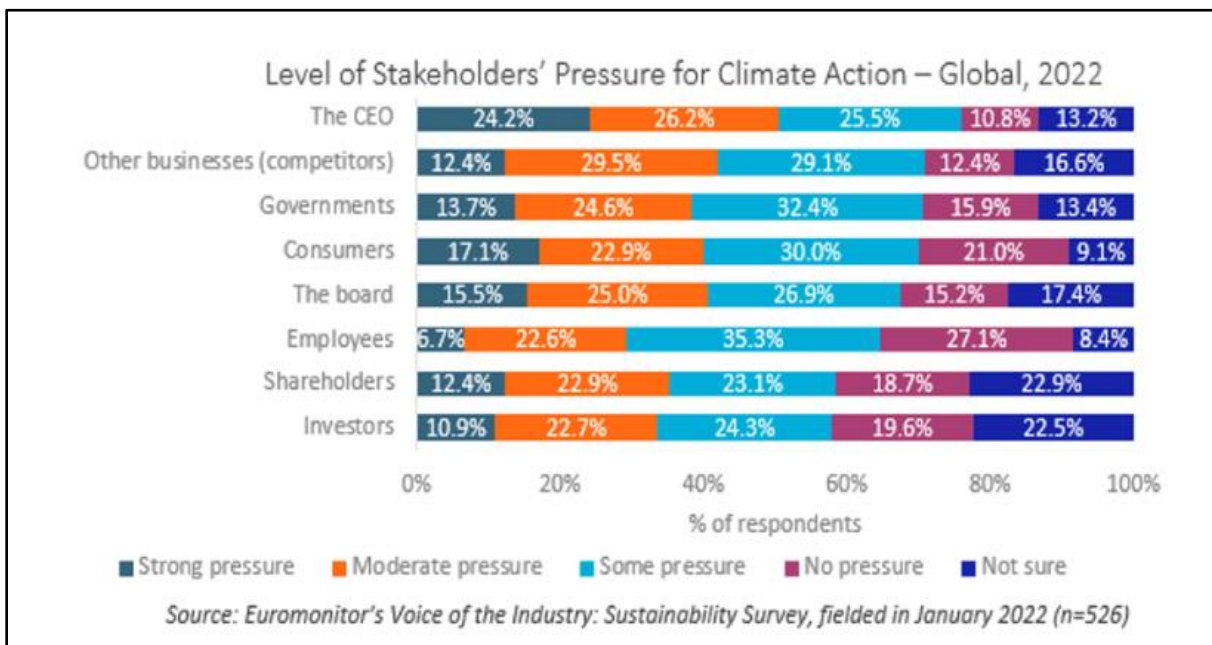


**Can sustainability-driven innovation create competitive advantages
that lead to superior financial performance?**

Research Paper By Vivana Contractor

Introduction

A sustainability survey by Euromonitor found that experts concur that climate change is changing consumer demand and will have an impact on the supply chain, with 76% and 71% of responses, respectively. In 2022, two out of three respondents to Euromonitor's Voice of the Consumer: Lifestyles survey stated that they were concerned about climate change, indicating that consumer awareness of the issue is rising at the same time. Corporates are under pressure to adopt more climate-friendly business practices as environmental concerns have a greater impact on company performance. This pressure comes from the CEO and rivals, but it also comes from governments and customers.



Since important stakeholders like investors and consumers closely heed the advice of thought leaders and science-based organizations regarding the need to act immediately, businesses must play a proactive role in assisting economies in transitioning to more sustainable models. In April 2022, for example, the Intergovernmental Panel on Climate Change (IPPC)

released a report stating that if the temperature trends of the Paris Agreement are to be adhered to, global emissions must peak by 2025.

Although 62% of global industry experts say that their company prioritizes ESG issues, most players don't follow through on their commitments and intentions, which far outweigh the actual actions taken. Global industry experts report that their companies only publish ESG reports at an average of 14%. The leading industry in ESG reporting is the food and beverage sector, with only 15% of respondents, three times higher than the last-place household essentials companies.

Businesses will face growing legal and reputational risks as a result of the growing discrepancy between what they say and do about ESG as consumers, investors, and regulators place a greater value on it. To mitigate these risks, it is imperative that businesses take significant action to live up to their rhetoric. This often calls for not only financial investment but also behavioral and attitude changes as well as openly tracking and disclosing results.

Demands for corporate and governmental action are becoming louder, and consumers are demanding sustainability. According to a Meta insights report released earlier this year, consumers are willing to pay more for sustainable or eco-friendly products and are seeking to identify with brands that reflect their beliefs. Customers are increasingly interacting with brands that post content about their sustainability efforts, according to research, which highlights the importance of purpose-driven messaging and products since the pandemic began. Although the generational gap is narrowing, Meta discovered that millennials are 1.3 times more likely than older generations to care about sustainability. With interest peaking in 2021, sustainability metrics like carbon neutrality are increasingly being discussed in society online

Sustainability is a means to differentiate and fortify a brand, and it's not just about doing the right thing. Prioritizing social and environmental impact helps businesses gain trust, which is crucial in the competitive market of today. Companies can stand out and develop stronger relationships with stakeholders by being transparent and taking deliberate action. With the recent US presidential election, it is no coincidence that 80 percent of companies (34% maintain or 44% increase) plan to maintain or increase their sustainability investments, according to the most recent PwC Pulse Report.

Resilience is based on sustainability. Businesses that make sustainable investments, from energy use to supply chains, are better able to withstand setbacks and emerge stronger. In an uncertain world, these initiatives not only reduce risks but also put companies in a position to take advantage of new opportunities. The Sustainability for Business Resilience Report 2024, for example, emphasizes how incorporating sustainable practices improves a business's capacity to foresee and respond to disruptions.

Brand strength is fueled by sustainability and purpose, which in turn creates long-term value. Brands that invest in these areas gain the trust of investors and draw in devoted consumers, according to the Brand Finance Global 500 2025 Report. They are strategies for long-term success rather than short-term gains.

Investors now insist on transparency regarding sustainability initiatives. The Connected Impact Transparency Index for 2024 demonstrates how businesses that match ESG disclosure with action increase capital and foster greater trust. Long-term value is becoming more and more important to investors than immediate profits, and purpose-driven businesses are turning out to be wise investments.

As employee expectations continue to evolve, it's clear that purpose plays a significant role in attracting and retaining top talent. According to the 2025 Edelman Trust Barometer, 63% of workers are concerned about how automation and globalization will affect their future. Businesses that incorporate sustainability into their culture encourage engagement and loyalty from their staff in addition to providing a sense of security. Meanwhile, two-thirds of workers believe their employers are not doing enough to address sustainability and climate change, according to a January Deloitte report.

Companies that share their values are rewarded by consumers, and this trend is only getting stronger. As per the recently published Brand Finance Global 500 2025 Report, the value of the top 500 brands worldwide grew by 10%, significantly surpassing the growth of the global economy. Customers, especially the younger ones, are searching for companies that do more than just say the right things when it comes to sustainability.

Conceptual Framework

The curation of completely new goods, services, or processes that provide noteworthy benefits for the environment and society in comparison to currently available options is known as sustainable-driven innovation (SI). Sustainable innovations aim to keep a fine balance between the welfare of the society, ecological unification, and economic growth, in comparison to innovations that are conventional, which may be basically focused on profitability or efficiency. The installation of technologies for renewable energy, like windmills and solar panels, is a prime example of SI. By giving us an alternative for fossil fuel and promoting the usage of clean energy, these advancements help in carbon emission reduction.

Crucial components showing branches of sustainable innovation,

Environmental Impact: alleviating unfavorable environmental impacts, such as pollution or the exhaustion of natural resources.

Economic viability: Ensuring if the invention can complement growth and is feasible commercially.

Social responsibility: Thinking about broader social consequences, like the health of the community, better working conditions, and global fairness.

There are certain forms of sustainable innovation:

Product innovation: it is the procedure of advancing new products that are sustainable or highly evolving the present ones. Electric vehicles (EVs) are an instance of a product innovation curated to lessen emissions of greenhouse gas.

Process Innovation: Businesses also utilize greener and cleaner manufacturing techniques to pioneer the ways in which their products are made. Integrating energy-efficient producing procedures or circular economy simulations, which cut back on waste through reusing materials and recycling, are some of the instances of process innovations that are sustainable.

Business Model Innovation: The foundation of the business model is combined with this kind of innovation, which integrates sustainability. The collaborative economy, in which enterprises such as Uber or Airbnb open avenues to services without having the ownership, is a suitable instance of how to reduce waste and consumption of resources. Several examples of SI can be discovered in various industries. One example is Patagonia, an outdoor apparel company well-known for its commitment towards environmental sustainability. Patagonia promotes environmental care and repair, in which the company has actually used materials that are recycled in its products for its Worn Wear initiative. Tesla, by creating electric cars that considerably reduce carbon emissions, has disrupted the automotive sector in the energy sector.

Over and above automobiles, its commitment to sustainability is shown by its solar energy products, which offer end-to-end solutions that are energy efficient. Another instance of SI is the Zero Waste movement, which has been welcomed by businesses such as Unilever with the aim of eliminating waste from their production processes, and is another illustration of SI.

Eco-innovation is an unconventional business strategy that not only advances a company's performance and competitiveness but also encourages sustainability over a product's total life cycle. This can be used by small and medium-sized businesses (SMEs) to penetrate new and growing markets, boost production, attract new capital, get a profitability boost throughout the value chain, and be one step ahead of rules and regulations, especially those related to the environment.

Green innovation is the execution of ideas that are innovative to keep the adverse effects of the environment at a minimum level and make sustainability effective. It entails creating new business models, procedures, goods, services, or technologies that are more ecologically friendly than the ones they replace. Through waste reduction or energy efficiency improvements, green innovations can also increase resource efficiency. We can guarantee that our natural resources are used responsibly and are available for future generations by coming up with creative ways to lessen our environmental impact.

The fundamental concept of radical innovation is the creation of an entirely new technology that can produce a product that has never been offered before, whereas incremental innovation strategy aims to utilize feedback from consumers or other stakeholders to create more appealing solutions that increase the profitability of current products.

An internal dimension is the first, depending on the resources and knowledge involved. An incremental innovation will improve competence by building on the resources and

knowledge already available to a particular organization. Conversely, a radical innovation will necessitate entirely new resources and/or knowledge, which will destroy competence. Using the impact on market competitiveness and technological advancements,

The second dimension, the external one, distinguishes the innovation. The current products on the market will continue to be competitive, while an incremental innovation entails small technological adjustments. Instead, a radical innovation would entail significant technological breakthroughs that would make the current products outdated and non-competitive. The vast and continuous process of developing strategies is referred to as strategic management. Putting decisions and actions that will direct an organization into practice and assessing them. Toward accomplishing its long-term goals and preserving its position in the market. The alignment of an organization's resources and capabilities is all part of this process. Actions in relation to its objectives and the surrounding environment.

The distinct edge a business has that enables it to is known as its competitive advantage. Outperform its competitors in delivering superior value to customers, which ultimately. Results in increased market share, profitability, or client retention. An organization succeeds. Competitive advantage through the creation of unique skills and creative solutions. Utilizing its resources in ways that are difficult for rivals to imitate.

According to Porter's competitive advantage theory, competitive advantages fall into various categories:

Cost Leadership: Using the cost leadership strategy, a business becomes the industry's lowest-cost producer. By attaining economies of scale, streamlining operations, and reducing expenses, the business can provide goods and services at a lower cost than rivals without compromising

profitability. In markets where consumers are price sensitive and look for the best deal without sacrificing quality, cost leadership works very well.

Differentiation: Offering distinctive goods or services that stand out in the market is the goal of differentiation. Businesses that use this approach make significant investments in design, quality, customer service, innovation, and branding. Businesses can charge more for their products because of differentiation because customers believe that they offer something special that rivals' products don't. This tactic works best in markets that value individuality and have a wide range of consumer preferences.

Focus (or Niche): Concentration entails focusing on a particular market segment, region, or clientele and creating customized goods or services to meet the particular requirements of that segment. Businesses using this proposition are in competition in the chosen niche either through price (cost focus) or differentiation of products (differentiation focus). These businesses can provide highly specialized services or goods that might not be appealing to the general public but command loyalty and higher margins within the niche by knowing the particular preferences of a targeted segment.

According to the Resource-Based View (RBV), a company's internal resources and capabilities are essential for creating a long-lasting competitive edge. Not all RBV places more emphasis on the external factors than other approaches that do. Significance of utilizing an organization's own distinctive assets to surpass rivals. These resources fall into three different categories.

Physical Resources: These comprise material possessions like tools and machinery.

Technologies as well as facilities. Better physical resources enable businesses to cut costs.

Lower expenses or boost the effectiveness of production.

Human Resources: Human capital, which encompasses the abilities, expertise, and inventiveness of employees, is essential to the growth of innovation and customer connections as well as operational efficiency.

Organizational Resources: These are the capabilities of the business in terms of routines, procedures, leadership, culture, and structure that are well-organized. Resources enable businesses to match their operations to their strategic objectives and quickly adapt to environmental changes.

RBV states that a firm's possession of resources gives it a competitive advantage, which are valuable, rare, inimitable, and organized. The VRIO framework encapsulates this.

Valuable: A resource must allow the business to take advantage of opportunities, eliminating market threats.

Rare: If rivals do not widely own a resource, it is considered rare, which is difficult to replicate or replace by others.

Inimitable: The resource must be expensive or difficult for rivals to copy. Often, this is due to factors such as historical circumstances, causal ambiguity (an unclear connection between competition and resources), or social complexity (unique ties and culture).

Organized: The business needs to be set up so it can function well using the resource to gain a competitive edge while making sure it is done correctly, with coordination and adherence to the plan.

A company can create a unique, rare, valuable, and well-organized resource by combining them. Long-term competitive edge that is challenging for rivals to match.

Long-term competitive advantage is largely fueled by sustainability and innovation, particularly as ethical and ecologically conscious business practices and consumer values grow.

Incorporating sustainability into business operations or creating novel products that tackle social or environmental concerns can help companies stand out in the marketplace and win over customers.

Sustainability: Businesses can lower expenses, protect themselves from regulatory risks, and attract eco-aware customers by investing in sustainable sourcing, renewable energy, and eco-friendly operations. In the long run, sustainable practices offer chances for cost savings in addition to improving reputation.

Innovation: Businesses are better able to adjust to shifting market conditions, maintain an advantage over rivals, and seize new opportunities when they consistently innovate their process, product, or business model. Innovation frequently opens up new markets and generates new sources of income.

Tesla is a prime example of how sustainability and innovation can provide a competitive edge. Being a leader in the automotive sector is a result of Tesla's emphasis on renewable energy sources and electric vehicle technology. Through occurring innovations in electric powertrains, capabilities that are self-driving, and battery technology, over and above meeting consumer demand, Tesla also captures market share in a forward-minded, highly optimistic industry.

Organizations can achieve success in the long term with the establishment of a unique market position and distinguish themselves from competitors by capitalizing on these sources of competitive advantage in a business atmosphere that is ever-changing and competitive.

How successfully an enterprise utilizes its resources to gain income and profit is known to be its financial performance. Balance sheets, income statements, and cash flow statements are some of the financial statements, which are used to evaluate financial performance. Some

important financial performance metrics are profitability, cash flow, revenue growth, return on assets (ROA), and return on equity (ROE).

While poor outcomes might be a signal of suboptimality or financial constraints, strong financial performance shows successful management and a good financial situation. Assessing financial performance is essential for stakeholders to make well-informed choices.

We can measure financial performance through several financial metrics, and it depends on multiple factors. For instance, we can see the financial statements of the company, as these are report cards for the company. All the information you require regarding its financial situation is provided by the cash flow, income, and balance sheets. We can also use financial ratios such as the current ratio, debt-to-equity ratio, and inventory turnover ratio, which help us compare financial patterns with other companies and also see patterns in performance.

Sustainability-driven Innovation and Financial Performance Through Competitive advantage

USERS	OBJECTIVES AIMED	Performance measurement indicators
Shareholders	Evaluating the value of the entity and the ability to remunerate the invested capital	- Net profit and rates built on it - Indicators that express value creation
Managers	Estimating the strategic and tactical objectives and the degree of their achievement	- Indicators of resource use (cost and expenditure indicators) - Indicators of result, efficiency, effectiveness - Activity indicators (management) - Indicators specific to sustainable development
Creditors	Determining the company's ability to generate cash or cash equivalents	- Liquidity indicators - Solvency indicators - Cash flow indicators
Employees	Establishing the form and level of remuneration, as well as assessing the stability of the enterprise	- Activity indicators - Efficiency indicators
Customers	Estimating the quality of production and assessing the stability of the entity	- Production quality indicators - Total quality indicators
Public	Appreciation of sustainable development	- Specific sustainable development indicators

Source: (Berheci, 2010, pg. 377)

In the above table we can see that in the context of measurement, performance can be measured using a variety of indicators, which stem from the realization that evaluating an entity's health solely on the basis of its profit is insufficient. These indicators include profit, results, sustainable development, liquidity, profitability, efficiency, and cash flow. These factors vary depending on the interests of users of accounting information.

Over time, business entities have understood the importance of the measurement of performance in a dynamic working environment. Apart from the fact that profits have always been taken into account when measuring financial performance, the performance of an entity is determined by considering numerous other factors and using varied methods of analysis.

The assessment of an entity's success is incomplete without considering factors such as sustainable development, liquidity, cash flows, efficiency, anomalies, and risks that may affect economic and business operations, and more, along with profitability. All these aspects need to be considered for the accurate substantiation of opinions.

Information about the financial performance of an entity is essential to consider when evaluating both the internal and the external environments. It is essential to include the expectations and interests of individuals involved in the social and economic life of the entity. Understanding an organization's performance can be done by using a system of measuring and managing performance, which collects and assesses information to support business decision-making processes. Performance measurement plays the role of diagnostics and monitoring of financial activities.

The state of profitability highlights the performance of an organization in two forms: absolute form, which refers to calculating profitability by calculating profit levels, and relative form which refers to calculating profitability by calculating profit rates. Measurements of

financial performance are heavily dependent on and are influenced by financial statements, in which the primary elements for assessing results and profitability are revenue and expenses. The performance of an economic unit is reflected through its earnings, whereas the performance of the entity is reflected through the profit and loss account.

Creating value for creditors, shareholders, workers, the government, and the economy is just as important to a company's performance as making a profit. Value added, which reflects a company's contribution to GDP and national wealth is the excess that it produces over what it consumes. It is thought to be a more significant indicator than turnover since it gauges how well resources are put to work creating wealth. The global financial crisis of 2008 made it clear that managers must concentrate on creating value in order to maximize shareholder wealth. Both additive and subtractive methods can be used to calculate value added, which is a tool for rewarding stakeholders while maintaining financial stability, productivity, and long-term growth.

The results of early ESG adopters are convincing, demonstrating that sustainability is not only a moral position but also a competitive advantage. Al Gore's Generation Investment Management, which was established in 2004, is among the most prominent instances. After initially being written off as a passing woke experiment, the company has proven to be profitable over the long term, which has resulted in the establishment of a \$1.07 billion sustainable solutions fund.

Companies that prioritize clean energy, circular fashion, and ethical production models are not just surviving but thriving, as demonstrated by companies such as Tesla and Patagonia. The financial sector is one that has profited immensely from ESG. Professionals' strategies have been bolstered by their knowledge of climate risks in financial institutions, sustainable finance, and impact investing. Thornburg Investment Management, a four-star firm, incorporates ESG

considerations into their financial analysis to identify businesses that meet the criteria for profitability and positive impact. Using their metrics, they were able to generate an impressive average annual return of 10.8% over a five-year period.

As it turns out, ESG is a competitive advantage that rewards progressive leadership with increased investments and creative approaches, not just a moral compass or defensive tactic. The model also naturally reduces operational, reputational, and regulatory vulnerabilities, which lowers long-term risks.

Empirical Evidence

We will now see the relationship between sustainability and financial performance internationally.

Sample and Data

Five emerging economies, Brazil, Russia, India, China, and South Africa, are represented in the sample, which consists of non-financial businesses from a variety of industries. The five years from 2014 to 2018 comprise the data period. The motivating element in the samples' composition was determined by the ESG scores' accessibility. Gathering sustainability data was the first stage, and gathering financial information for the businesses that had sustainability data was the second. The sample consists of the companies with data for the last five years, while the companies with any missing data are not included.

Country	No. of companies
Brazil	23
Russia	10
India	51
China	8
South Africa	17
<i>Total</i>	<i>109</i>

Variables

Under dependent variables we are taking factors that determine the financial performance of an organization. To analyze the effect of sustainability on financial performance, we have taken various financial performance metrics such as Return on Assets (ROA), Return on Equity (ROE), Net Profit Margin (NPM), and Operating Profit Margin (OPM). These ratios are measured by taking the reference of financial statements, which are curated according to International Financial Reporting Standards (IFRS) provided in US dollars. This will prevent the problem of variations that may occur due to national accounting treatments, enabling intercountry comparisons.

Under independent variables we have taken sustainability performance metrics. For this we are taking ESG scores. ESG stands for Environment, Social, and Governance. There are a total of four scores consisting of the total ESG score and a score each for the different components of the total ESG score. Sustainalytics is the source for the scores, which are from the top independent global provider of ESG and corporate governance research and ratings to investors.

In the models, we eliminated the variables by using a number of control variables. Effects of other elements and determining how financial performance is affected by sustainability. Such control variables are leverage, size, GDP growth rate, and the Human Development Index (HDI). Leverage is measured by the help of the ratio of total debt to total assets and will help to track the company's financial risk level. The effect of company size is revealed by utilizing the natural logarithm of total assets as a means to measure size. The gross domestic product (GDP) growth rate is a mirror of the country's economic development as a whole and is estimated as the change in GDP from year to year. Life expectancy, education, and per capita income are all taken into

account by the United Nations Development Programme (UNDP), which takes care of the life expectancy, per capita income, and education while estimating its Human Development Index. In order to account for the impact of those national factors, HDI is included as a control variable.

Methodology and Models

Stakeholder theory forms the foundation of this study's methodology. According to the stakeholder theory, a company must satisfy the demands and expectations of both internal and external stakeholders. ESG practices are recognized as environmental, social, and governance practices. The involvement of stakeholders to meet those expectations, which ultimately results in improved financial performance. As a result, it is anticipated that sustainability performance and financial performance will positively correlate. The study's dataset, a panel dataset, includes time series and cross-sectional data for the sample businesses.

To determine the effect of sustainability performance on financial performance, we created a number of models; the dependent variable is a financial performance ratio, the independent variables are the total ESG score or the component scores, and panel regressions are performed. There are two types of groups. Group 1 uses the total ESG score, and Group 2 uses environmental, social, and governance scores as independent variables.

Group 1. The models are listed here:

$$\text{Model 1: ROA} = \beta_0 + \beta_1 \text{TESG}_{i,t} + \beta_2 \text{LEV}_{i,t} + \beta_3 \text{SIZE}_{i,t} + \beta_4 \text{GDPGR}_{i,t} + \beta_5 \text{HDI}_{i,t} + \varepsilon_{i,t}$$

$$\text{Model 2: ROE} = \beta_0 + \beta_1 \text{TESG}_{i,t} + \beta_2 \text{LEV}_{i,t} + \beta_3 \text{SIZE}_{i,t} + \beta_4 \text{GDPGR}_{i,t} + \beta_5 \text{HDI}_{i,t} + \varepsilon_{i,t}$$

$$\text{Model 3: OPM} = \beta_0 + \beta_1 \text{TESG}_{i,t} + \beta_2 \text{LEV}_{i,t} + \beta_3 \text{SIZE}_{i,t} + \beta_4 \text{GDPGR}_{i,t} + \beta_5 \text{HDI}_{i,t} + \varepsilon_{i,t}$$

$$\text{Model 4: NPM} = \beta_0 + \beta_1 \text{TESG}_{i,t} + \beta_2 \text{LEV}_{i,t} + \beta_3 \text{SIZE}_{i,t} + \beta_4 \text{GDPGR}_{i,t} + \beta_5 \text{HDI}_{i,t} + \varepsilon_{i,t}$$

Group 2.

$$\text{Model 5: ROA} = \beta_0 + \beta_1\text{ENVS}_{i,t} + \beta_2\text{SOCS}_{i,t} + \beta_3\text{GOVS}_{i,t} + \beta_4\text{LEV}_{i,t} + \beta_5\text{SIZE}_{i,t} + \beta_6\text{GDPGR}_{i,t} + \beta_7\text{HDI}_{i,t} + \varepsilon_{i,t}$$

$$\text{Model 6: ROE} = \beta_0 + \beta_1\text{ENVS}_{i,t} + \beta_2\text{SOCS}_{i,t} + \beta_3\text{GOVS}_{i,t} + \beta_4\text{LEV}_{i,t} + \beta_5\text{SIZE}_{i,t} + \beta_6\text{GDPGR}_{i,t} + \beta_7\text{HDI}_{i,t} + \varepsilon_{i,t}$$

$$\text{Model 7: OPM} = \beta_0 + \beta_1\text{ENVS}_{i,t} + \beta_2\text{SOCS}_{i,t} + \beta_3\text{GOVS}_{i,t} + \beta_4\text{LEV}_{i,t} + \beta_5\text{SIZE}_{i,t} + \beta_6\text{GDPGR}_{i,t} + \beta_7\text{HDI}_{i,t} + \varepsilon_{i,t}$$

$$\text{Model 8: NPM} = \beta_0 + \beta_1\text{ENVS}_{i,t} + \beta_2\text{SOCS}_{i,t} + \beta_3\text{GOVS}_{i,t} + \beta_4\text{LEV}_{i,t} + \beta_5\text{SIZE}_{i,t} + \beta_6\text{GDPGR}_{i,t} + \beta_7\text{HDI}_{i,t} + \varepsilon_{i,t}$$

Where;

ROA: Return on Assets, ROE: Return on Equity, OPM: Operating Profit Margin, NPM: Net

Profit Margin, TESH: Total ESG Score, ENVS: Environmental Score, SOCS: Social Score,

GOVS: Governance Score, LEV: Leverage, SIZE: Firm Size, GDPGR: Gross Domestic Product

Growth Rate, HDI: Human Development Index

Results and Findings**Descriptive Statistics:**

Variable	Mean	SD	Min	Max	Skewness	Kurtosis
TESG	61.51	10.99	40	95	0.483	2.664
ENVS	61.14	13.31	31	100	0.334	2.766
SOCS	59.22	12.70	31.49	98.18	0.383	2.728
GOVS	66.38	11.93	37	100	0.169	2.595
ROA	0.08	0.08	-0.31	0.32	0.028	5.004
ROE	0.16	0.25	-2.55	2.94	0.060	59.385
NPM	0.11	0.15	-0.55	1.85	4.393	53.260
OPM	0.17	0.13	-0.52	0.69	-0.061	5.738
LEV	0.52	0.18	0.02	0.98	0.022	2.403
SIZE	23.34	1.35	19.36	26.46	-0.231	3.021
GDPGR	0.04	0.04	-0.04	0.08	-0.521	1.871
HDI	0.70	0.07	0.62	0.83	0.383	1.821

Note(s): No. of Observations: 545

The above table presents the descriptive statistics of the variables. Out of 100, the overall ESG score and its constituent environmental, social, and governance scores are calculated. The samples' T ESG score ranges from 40 to 95, with an average of 61.51. The average values for the components fall between 59 and 66. According to these scores, the sample companies' overall sustainability performance is mediocre. ROA has an average of 8%, a minimum of 31%, and a maximum of 32% in relation to the financial performance variables. ROE has a minimum of 255% and a maximum of 294%, with an average of 16%.

Since ROE is partially influenced by the capital structure of the companies, there is more variation in it; companies with a lower equity level may have higher ROE values. NPM ranges from 55% to 185%, with an average of 11%. OPM has a 52% minimum, a 69% maximum, and an average of 17%. The fact that OPM only reports operating profit, whereas NPM includes all operating and non-operating items, explains why NPM is more variable. With a minimum of 2% and a maximum of 98%, leverage (LEV) has an average of 52%.

The capital structure of the sample is generally equally balanced, despite the fact that some companies are almost entirely financed by debt or almost entirely by equity. With an average of 23.34, a minimum of 19.36, and a maximum of 26.46, SIZE, which is calculated as the natural logarithm of total assets, shows little variation in sample company size. During the study period, the countries' GDP growth rates range from 4% to 8%. Another country-level variable is the HDI, which ranges from a minimum of 62% to a maximum of 83%, with an average of 70%.

Pairwise Correlations:

	ENVS	SOCS	GOVS
ENVS	1		
SOCS	0.643*	1	
GOVS	0.522*	0.578*	1
ROA	0.194*	0.083	0.085*
ROE	0.182*	0.096*	0.086*
NPM	0.029	-0.009	0.018
OPM	0.047	0.043	0.039
LEV	-0.045	0.044	-0.006
SIZE	-0.071	-0.128*	-0.099*
GDPGR	0.000	-0.162*	-0.200*
HDI	-0.120	-0.005*	-0.038

Note(s): *: significant correlations at 5%

	TESG	ROA	ROE	NPM	OPM	LEV	SIZE	GDPGR	HDI
TESG	1								
ROA	0.143*	1							
ROE	0.147*	0.673*	1						
NPM	0.002	0.622*	0.415*	1					
OPM	0.040	0.506*	0.340*	0.592*	1				
LEV	0.0007	-0.443*	-0.028	-0.303*	-0.260*	1			
SIZE	-0.114*	-0.391*	-0.173*	-0.055	-0.031	0.344*	1		
GDPGR	-0.127*	0.238*	0.061	0.014	-0.047	-0.141*	-0.284*	1	
HDI	-0.066	-0.237*	-0.093*	-0.008	0.029	0.091*	0.318*	-0.774*	1

Note(s): *: significant correlations at 5%

The above two tables show the pairwise correlations among the variables. They are treated separately because the TESG score is highly correlated to its components. This is due to the fact that TESG is dependent on the scores of its components.

Panel Regression Results:

Models	5) ROA		6) ROE		7) OPM		8) NPM	
Intercept	0.3676***	(0.000)	1.271*	(0.065)	0.0516	(0.816)	-1.266	(0.326)
ENVS	0.0007	(0.113)	0.003	(0.130)	-0.0002	(0.756)	0.0003	(0.777)
SOCS	0.0005	(0.146)	0.004***	(0.019)	0.0012*	(0.062)	0.0012	(0.201)
GOVS	0.0001	(0.723)	0.003*	(0.091)	-0.00007	(0.918)	0.0005	(0.747)
LEV	-0.1675***	(0.000)	-0.108	(0.494)	-0.1914***	(0.000)	-0.4657***	(0.005)
SIZE	-0.0131***	(0.002)	-0.116*	(0.091)	0.0039	(0.593)	0.0865	(0.318)
GDPGR	0.3144***	(0.023)	0.057	(0.953)	0.2259	(0.529)	0.7357	(0.131)
HDI	0.0168	(0.858)	1.566	(0.401)	0.0862	(0.694)	-0.7985	(0.485)
R ²	0.3717		0.0633		0.0797		0.0955	
Adjusted R ²	0.3027		0.0511		0.0678		0.0837	
Hausman test	10.37 (0.1688)		14.13 (0.049)		9.70 (0.2063)		16.51 (0.0208)	
Prob	0.0000***		0.0458**		0.0006***		0.0004***	
Observations	545		545		545		545	

Note(s): *, **, ***, significant at 1%, 5%, 10%, respectively

The above table shows the results for panel regression for the models in group 1. Here, we took total ESG as an independent variable. We used four profitability ratios as dependent variables, which are ROA, ROE, OPM, and NPM, in this group of models. The total ESG score significantly influences the profitability metrics of ROA and ROE, according to the panel regression results in the above table; however, OPM and NPM yield negligible results. Prob(F) illustrates the model's overall fit, which is negligible. The range of R^2 values is 9.7 to 29.15%.

Leverage, size, and GDP growth rate are significant in the first model, but the Human Development Index is not. SIZE is the sole significant control variable in the second model. For the third and fourth models it was discovered that leverage is a significant coefficient. The significant results for leverage show negative coefficients, suggesting that the profitability ratios of the companies with less leverage would be higher. The significant results for size also show negative coefficients, suggesting that smaller businesses would perform better financially. The Hausman test, which is used to compare the fixed-effects and random-effects models, revealed that model 4 had fixed effects while models 1, 2, and 3 had random effects.

Models	1) ROA		2) ROE		3) OPM		4) NPM	
Intercept	0.3591***	(0.000)	0.5914**	(0.050)	0.1187	0.568	-1.1175	0.338
TESG	0.0013***	(0.003)	0.0040***	(0.016)	0.0006	0.488	0.0011	0.504
LEV	-0.169***	(0.000)	0.0073	(0.960)	-0.1866***	(0.000)	-0.4645***	(0.004)
SIZE	-0.0129***	(0.001)	-0.0297***	(0.003)	0.0025	0.733	0.0838	0.324
GDPGR	0.3342**	(0.019)	0.2211	(0.701)	0.1529	0.659	0.6933	0.151
HDI	0.0261	(0.793)	0.0082	(0.979)	0.0637	0.766	-0.8225	0.496
R^2	0.2915		0.0470		0.0676		0.0971	
Adjusted R^2	0.2872		0.0399		0.0699		0.0887	
Hausman test	6.40 (0.2695)		9.05 (0.1072)		9.39 (0.0945)		15.52 (0.0084)	
Prob (F)	0.0000***		0.0053***		0.0006***		0.0002***	
Observations	545		545		545		545	

Note(s): *, **, ***, significant at 1%, 5%, 10%, respectively. *p*-values are reported in the parentheses

The above table shows the results for panel regression for group 2. Here, we took ENV5, SOCS, and GOVS as independent variables and four profitability ratios, namely ROA, ROE, OPM, and NPM, as dependent variables. The models in Group 2 have a significant overall fit (Prob(F)), just like the models in Group 1. R^2 values range from 9.55% to 37.17%.

With a few notable exceptions, the findings demonstrated that the profitability ratios are not significantly impacted by the components of ESG scores. While size, leverage, GDP growth rate, and control variables have significant coefficients, model 5's ESG score components all yielded insignificant results. The only significant coefficient in model 6 is size, and SOCS and GOVS are significant at the 1 and 10% levels, respectively. Only SOCS yielded noteworthy outcomes in model 7. Every element of the ESG scores in model 8 has an insignificant coefficient. In models 7 and 8, leverage is the sole discernible control variable. Leverage and size have negative coefficients of significant results, which is consistent with group 1 findings.

Summarizing Findings:

The purpose of this article was to examine whether ESG factors significantly impact a company's financial performance. The first group of models we run uses the overall ESG score as the independent variable, and the second group uses the individual scores, the environmental, social, and governance components of the ESG score, as the independent variables. The four profitability ratios, ROA, ROE, OPM, and NPM, were employed as the dependent variables in both groups. Overall, the results show that the ESG score as a whole produced largely significant results; individual scores showed insignificant results but with a few exceptions.

The components of the overall ESG score are shown by these results. A company's financial performance is influenced by its governance, social, and environmental performance. The findings therefore have a practical application for the businesses, as they should actively

participate in all aspects of sustainability. ROA and ROE are significantly positively impacted by the total ESG score in group 1, but OPM and NPM results are not statistically significant. The fact that OPM and NPM are profitability ratios derived from the profit/loss statement, whereas ROA and ROE are profitability ratios derived from the statement of financial position, may help to explain this.

In reference to the control variables, size and leverage are found to be significant with negative coefficients in the majority of the models; these findings suggest that the sustainability-financial performance relationship is negatively impacted by those variables. The human development index, GDP growth rate, and country-level control variables, with a few exceptions, yielded negligible results in the models. The conclusion is that the overall ESG score significantly improves. Financial performance suggests that companies should have sound corporate governance procedures, be more sensitive to social issues, and be aware of environmental concerns.

Case Studies

Volvo Updates Sustainability Strategy

Significant progress has been made by Volvo in reaching its climate action goals since the company's 2019 release of its sustainability strategy. Climate-neutral energy now powers 69% of Volvo's operations, up from 55% in 2019. Volvo's manufacturing facilities worldwide now use 100% climate-neutral electricity, up from 80% in 2019. Furthermore, since 2018, Volvo has lowered its CO₂ emissions per vehicle by 19%. Volvo has revised its 2030 sustainability goals to include the following. Cut CO₂ emissions per vehicle by 75% (in comparison with 2018 levels).

Compared to 2018 levels, cut the average car's operating energy consumption by 40%. Aim for an average of 30% recycled content throughout the fleet, with new car models having at least 35% recycled content. Cut water consumption in owned operations by 50% on average per vehicle (compared to 2018 levels). Reuse or recycling should account for at least 99% of all operational waste. By 2040, Volvo wants to have zero greenhouse gas emissions. The organization's intention to use carbon removals only to offset any emissions that cannot be avoided is explained by this goal, which builds on its prior objective of becoming climate-neutral by 2040.

Prioritizing the reduction of actual emissions over carbon removals is still Volvo's top priority, and it encourages its suppliers to follow suit. Furthermore, since 2019, Volvo has been committed to adopting the circular economy. About 25% of the aluminum in Volvo's EX30 small SUV is recycled, and roughly 17% of the steel and plastic in the vehicle is recycled. Volvo wants its fleet to use 30% recycled content on average by 2030, and new models of cars released after that year must have at least 35% recycled content. In addition, Volvo has set a target of reusing or recycling 99% of all waste by 2030, up from 94% of global production waste in 2022.

The Home Depot Redesigns Packaging

Global home improvement retailer The Home Depot has removed polyvinyl chloride plastic film and expanded polystyrene foam, like Styrofoam, from the packaging of its private-label goods. Both materials are currently challenging to recycle and may take more than ten years to decompose. The United States Plastics Pact, which directs corporate commitments to phase out plastic packaging, lists both as unnecessary or problematic materials.

Regarding plastics, Home Depot is taking the following actions. Redesigning packaging: From 2017 to 2023, Home Depot must redesign nearly 1,300 packages for its own brands, which

include lightbulbs, bathroom vanity counters, and ceiling fans. Styrofoam reductions: The overhaul reduced 39 million square feet of plastic film, which could cover 513 soccer fields, and 6 million cubic feet of Styrofoam, which is enough to fill 67 Olympic swimming pools, for 2023 alone.

Because of the redesign, the packages are now lighter and more compact, allowing for more items to be loaded onto trucks or put on shelves. By January 2028, Home Depot plans to switch to fiber-based packaging that is recyclable, compostable, or composed of recycled materials. This will result in a significant switch to corrugated blister cards, paper and paperboard, and molded pulp. Between its own operations and the use of products sold in its stores, Home Depot has pledged to reduce its greenhouse gas emissions by 42%. Scope 1 and Scope 2 emissions, which include stores and energy purchases, increased by roughly 5% each in fiscal 2023, according to the company's 2023 ESG report.

Pizza Hut Reduces Greenhouse Gas Emissions

The second-biggest pizza restaurant chain in the world, Pizza Hut, cut its greenhouse gas emissions from the mozzarella cheese it uses in its dishes by 10% in 2023. With over 20,000 locations at the end of 2023, the company attributes the decrease to its choice to buy at least half of its milk from a 10,000-farm cooperative that is dedicated to using practices that will reduce CO₂ and methane emissions by 30% by 2030. In order to meet sourcing targets by 2025, Pizza Hut announced its partnership with Dairy Farmers of America in August 2022.

The company achieved that goal more than a year ahead of schedule, according to Yum! Brands' 2023 sustainability report, which was made public on October 8. The cooperative Dairy Farmers of America currently provides 60% of the milk used to make cheese in the United States. That amounts to roughly 235 billion pounds of milk from 140,000 cows. The Habit

Burger & Grill, KFC, and Taco Bell restaurant chains are all owned by Yum!. Dairy made up 13% of Yums' overall Scope 3 supplier emissions in 2023. The carbon footprint of the company is 99% represented by scope 3.

About 22% of US emissions associated with food and agriculture are attributable to the dairy industry. Methane from cow burps and manure, as well as CO₂ released during feed production, are the two main causes for concern. For this reason, more food companies are paying attention to dairy emissions. Since dairy is one of the top five impacts for that division, Mars is spending \$47 million over the next three years to lessen the impact of the milk and butter used for its snack business. Reducing dairy emissions is a key component of General Mills' 2030 reduction targets, which it achieved by utilizing the same benchmarking program as Pizza Hut.

Bohinj Eco Hotel's Embedded Sustainability

Embedded sustainability is the integration of environmental, health, and social values into the company's core operations with the aim of pursuing sustainable value for a variety of stakeholders, including suppliers, customers, employees, NGOs, and regulators, with whom the sustainable solutions for system-level changes are co-developed. The first fully eco-hotel in Eastern Europe, the Bohinj Eco Hotel, is a superb illustration of such a business. Slovenia's first and only Green Globe-certified hotel, the Bohinj Eco Hotel, is regarded as one of the world's top sustainable hotels.

The hotel has integrated environmental thinking and performance into every aspect of its operations, rather than concentrating on ancillary environmental features like standard eco-efficiency procedures. With a combination of cogeneration and geothermal technologies, the hotel generates its own energy for its aquapark and all other hotel operations. Heat and water are continuously recycled. Combining energy-efficient LED lighting with wall and window

insulation enables the highest levels of comfort at affordable prices. Standard hotels in the area produce 174.82 kg of CO₂ per night, while the hotel produces 17.22 kg per guest. The hotel is able to provide better service without raising prices by reinvesting the energy cost savings into other operations like food and catering.

H&M's Sustainability via Collaboration and Prioritizing Social Goals

Handamp;Ms initiatives in Bangladesh serve as an example of how industry-wide cooperation is necessary to address sustainability issues, such as equitable wages, rather than taking discrete action. In order to promote living wages through collective bargaining, H&M formed ACT (Action, Collaboration, Transformation) in 2015 with Industry ALL and 22 international brands. This was done because the company does not directly control wages or own factories. This was the first significant cooperative effort in the apparel industry to deal with wage concerns as a group. The case demonstrates how companies implementing sustainable business models redefine success as resolving environmental or social issues collaboratively rather than solely focusing on financial gain. These companies prioritize the creation of value for stakeholders over shareholder profit in their pursuit of cooperative advantage.

This change calls into question the conventional wisdom regarding competitive advantage and encourages companies to embrace the triple bottom line (people, planet, profit), which acknowledges that sustainability efforts can have a significant positive social and environmental impact even if they result in longer-term financial gains. In the end, achieving a sustainability-driven advantage necessitates reevaluating competition with an emphasis on cooperation, stakeholder involvement, and more extensive sustainable value generation.

H&M is a fitting example of sustainability in collaboration, executing its circular, climate-friendly, and equilibrium strategy of fashion all across a large-scale global supply chain.

It has been posited as the most sustainable consumer goods company in Sweden and has attained high ratings for effective supply chain management, human rights, and reporting even after top rankings in the Dow Jones Sustainability Index. Thereafter, H&M has consistently been ranked in the Top 50 Global Brands and Retailers, signalling its maintained competitiveness and dedication to sustainability. Nevertheless, the business has been criticized for some ongoing disputes regarding substandard working conditions, low pay scale, and child labor, and, not long ago, forced labor was claimed in Xinjiang. Notwithstanding the frequent application of remedial actions, risks to reputation will still exist. The conflict between Russia and Ukraine in 2022 revealed the susceptibility of sustainability-based advantage.

Approximately \$190 million in anticipated revenue was lost by H&M, and it saw a 10% dip in profits when it shut down 150 of its Russian stores, which was 4% of its total sales. This case study exhibits how geopolitical crises and foreign shocks can cut off a company's competitive advantage through sustainability, leading it to rethink its goals and settle for financial and social performance that is lower even in the presence of robust sustainability credentials.

Challenges and Opportunities

Sustainability is becoming even more prevalent among businesses as a source of innovation providing competitive advantage and is merely not just a moral or compliance requirement. In exchange for perceiving social and environmental pressures as hindrances, forward-looking companies are inculcating sustainability into their operations and strategies, leveraging them into avenues for resilience and growth. Incorporating innovative concepts, technologies, and business models that create an equilibrium between financial liquidity and acceptable social and environmental consequences is known as sustainable innovation. This

consists of actions like optimizing allocation of resources, recycling and reusing, incorporating renewable energy, and inculcating principles of sustainability into a company's daily operations. In this way, innovation furnishes both long-term value to the society and financial wellness. Sustainability-driven innovation has many hindrances even though it has potential. High initial costs and discouraging investment due to delayed financial returns make the measurement of sustainability impact difficult. More obstacles such as competition from sectors with fully built conventional models, trade-offs between sustainability and financial plans, and the issue of handling the claims of several stakeholders. For many businesses adoption is also risky due to no fixed results of radical processes like circular economy models.

Simultaneously, sustainable innovation provides a lot of possibilities to gain advantage. Businesses that utilize it can bring in new products or business models to increase their competitiveness and be economical by enhancing efficiency. Thereafter, sustainability gives a boost to the company's reputation, giving it a head start in cutting-edge technologies, and opens pathways to undiscovered markets and affiliations.

In certain circumstances, it even allows for making breakthroughs in discoveries that overturn entire sectors and disrupt how value is created. To conclude, there are both opportunities and challenges related to sustainable-driven business innovation. Although the operational, financial, and cultural obstacles are there, which businesses have to overcome, it has humongous potential to supply long-term growth, resilience, and value to the stakeholder. Businesses that fruitfully incorporate sustainability into their strategies for innovation have more chances to secure sustaining competitive advantages in an ever-evolving world economy.

Conclusion

Sustainability is currently a main ingredient in the long run for achieving corporate success. Companies that entirely implement sustainability and ESG principles into their agendas and decision-making processes seem to have a competitive advantage, enhanced resilience, and better monetary stability. Today's economy is getting more and more affected by social and environmental demands; companies that see sustainability as an acceptance exercise have the risk of falling short of their competitors.

Implementing sustainability into governance, finance, culture, and operations reflects going beyond reporting and acceptance. Leadership qualities are essential, with stakeholders and board members taking responsibilities and giving incentives to ESG objectives. Decisions about money allocation and investments, which go hand in hand with laid-down economy programs and dependable financing, must be in tune with sustainability predominance.

However, businesses must implement ESG risks, like supply chain risks, regulatory challenges, and climate change, in their main risk management structure. Inter-team coordination and the implementation of sustainability into monetary planning, R&D, procurement, and daily operations are required for a perfect incorporation. This requires the setup of the business to be one that upholds ESG regulations with training, bonuses, and communication imultaneously, a team's support and reliability are enhanced by creating transparent and responsible brands that meet customer benchmarks. This path has clear business advantages, including enhanced roadmaps to green finance, better placement in places of strict regulations, better financial scalability through efficiency gains, and increased brand confidence. In addition, companies that frequently implement sustainability enhance their resistance to obstacles and pave new growth prospects in creating scalable markets.

Finally, it should be noted that incorporating sustainability into business decision-making is now necessary for survival and long-term competitiveness. Leadership dedication, integrated finance and governance, proactive risk management, and an organizational culture of sustainability are all necessary for success. Businesses that adapt to this change will be in the best position to take the lead in moving toward a profitable, sustainable future.

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